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Department of Mathematics

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Math 208 Major Exam I Term 242

Original exam date: 18 February 2025. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. If the vectors $\mathbf{u} = (3, -4, 5)$, $\mathbf{v} = (1, -2, 3)$ and $\mathbf{w} = (6, -8, a)$ are linearly dependent, then $a =$

- (a) 7
- (b) 10
- (c) 11
- (d) 8
- (e) 9

2. The general solution of the differential equation $x^2y'' + 2xy' = 12x^3$ is $y = (A$ and B are real numbers)

- (a) $x^2 - \frac{A}{x^2} + B$
- (b) $x^3 - \frac{A}{x} + B$
- (c) $x^3 - \frac{A}{x^2} + B$
- (d) $x^2 + \frac{A}{x} + B$
- (e) $x^3 - \frac{A}{x} + Bx$

3. Let $\mathbf{u} = (2, 7)$, $\mathbf{v} = (3, 8)$ and $\mathbf{t} = (0, 5)$. If $\mathbf{t} = a\mathbf{u} + b\mathbf{v}$, then $a + b =$

- (a) 0
- (b) -2
- (c) -1
- (d) 2
- (e) 1

4. When sugar is dissolved in water, the amount A that remains undissolved after t minutes satisfies $\frac{dA}{dt} = -kA$ ($k > 0$). If 25% of the sugar dissolves after 1 minute, how long does it take for half of the sugar to dissolve (in minutes)?

(a) $\ln(4/3)$

(b) $\frac{1}{\ln(7/3)}$

(c) $\frac{\ln 2}{\ln(7/4)}$

(d) $\frac{-1}{\ln(3/4)}$

(e) $\frac{\ln 2}{\ln(4/3)}$

5. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

(a) The set of all vectors (x_1, x_2, x_3, x_4) such that all x_i 's are nonzero

(b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 3x_3$ and $x_2 = 4x_4$

(c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 + x_3 + x_4 = 5$

(d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_2 = x_3x_4$

(e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_2x_3x_4 = 0$

6. The solution space of the system

$$x_1 - 3x_2 - 7x_3 - 10x_4 = 0$$

$$x_1 + x_2 + 9x_3 + 2x_4 = 0$$

$$x_1 - 2x_2 - 3x_3 - 7x_4 = 0$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s and t are real numbers and

(a) $\mathbf{u} = (-5, -4, 1, 0)$, $\mathbf{v} = (1, -3, 0, 1)$

(b) $\mathbf{u} = (-5, 4, 1, 0)$, $\mathbf{v} = (1, -3, 0, 1)$

(c) $\mathbf{u} = (-5, -4, 1, 0)$, $\mathbf{v} = (1, -3, 0, -1)$

(d) $\mathbf{u} = (-5, -4, 1, 0)$, $\mathbf{v} = (1, 3, 0, 1)$

(e) $\mathbf{u} = (5, -4, 1, 0)$, $\mathbf{v} = (1, -3, 0, 1)$

7. If the acceleration of a particle is $a(t) = \frac{1}{(t+1)^3}$, the initial velocity is $v_0 = 0$, and the initial position is $x_0 = 0$, then $x(1) =$

- (a) $\frac{1}{4}$
- (b) 1
- (c) $\frac{1}{8}$
- (d) $\frac{1}{2}$
- (e) $\frac{1}{16}$

8. If we substitute $y = e^{rx}$ into the differential equation $5y'' - 6y' - y = 0$, then the product of all values of r is

- (a) $-\frac{1}{6}$
- (b) 0
- (c) $\frac{1}{6}$
- (d) $\frac{1}{5}$
- (e) $-\frac{1}{5}$

9. The solution of the initial-value problem $\frac{dy}{dx} = y^2 \sin x - y^2 \cos x$, $y\left(\frac{\pi}{2}\right) = \frac{1}{3}$ is

- (a) $y = \frac{1}{\sin x + \cos x + 2}$
- (b) $y = \frac{1}{-\sin x + \cos x + 4}$
- (c) $y = \frac{1}{\sin x + \cos x}$
- (d) $y = \frac{1}{\sin x - \cos x + 2}$
- (e) $y = \frac{1}{\sin x + \cos x - 1}$

10. If the differential equation $(6x^2 + kxy) dx + (x^2 - 6y) dy = 0$ is an exact differential equation, then $k =$

- (a) 1
- (b) -2
- (c) 0
- (d) -1
- (e) 2

11. A general solution of the exact differential equation $(1 + ye^{xy}) dx + (2y + xe^{xy}) dy = 0$ (where c is constant) is

- (a) $x^2 + e^{xy} + y^2 = c$
- (b) $x + e^{xy} = c$
- (c) $x + e^{xy} + y = c$
- (d) $e^{xy} + y = c$
- (e) $x + e^{xy} + y^2 = c$

12. A general solution of the homogeneous differential equation $2xy \frac{dy}{dx} = 4x^2 + 3y^2$ is (where c is a constant)

- (a) $y^2 - 4x^2 = cx^3$
- (b) $y^2 + 2x^2 = cx^3$
- (c) $y^2 + 4x^2 = cx$
- (d) $y^2 - 2x^2 = cx^3$
- (e) $y^2 + 4x^2 = cx^3$

13. The general solution of the differential equation $2xy' - 3y = 9x^3$ is (where c is a constant)

- (a) $y = 2x^3 + cx^{2/3}$
- (b) $y = 3x^3 + cx^{2/3}$
- (c) $y = x^3 + cx^{3/2}$
- (d) $y = 2x^3 + cx^{3/2}$
- (e) $y = 3x^3 + cx^{3/2}$

14. Given that $y = \frac{1}{c-x}$ is a family of solutions of $\frac{dy}{dx} = y^2$, the largest interval of existence of the solution of the initial-value problem $\frac{dy}{dx} = y^2$, $y(1) = 2$ is

- (a) $(\frac{3}{2}, \infty)$
- (b) $(-\infty, 2)$
- (c) $(-\infty, \frac{3}{2})$
- (d) $(-\frac{3}{2}, \infty)$
- (e) $(-\infty, 3)$

15. By using a suitable substitution, we can transform the differential equation $3xy^2y' = 3x^4 + y^3$ into the linear differential equation

(a) $v' + \frac{1}{x}v = x^2$

(b) $v' - \frac{1}{x}v = 3x^3$

(c) $v' + \frac{1}{x}v = 3x^3$

(d) $v' - \frac{1}{x}v = x^3$

(e) $v' + \frac{1}{x}v = 3x^2$

Answer Key

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Q	Ans	Brief explanation
1	(b)	$\det = -2a + 20 = 0$, so $a = 10$ (then $\mathbf{w} = 2\mathbf{u}$).
2	(b)	$p = y'$: $(x^2p)' = 12x^3$, $p = 3x^2 + Ax^{-2}$, $y = x^3 - \frac{A}{x} + B$.
3	(e)	$2a + 3b = 0$, $7a + 8b = 5$ give $a = 3$, $b = -2$, so $a + b = 1$.
4	(e)	$e^{-k} = \frac{3}{4}$ so $k = \ln \frac{4}{3}$; $e^{-kt} = \frac{1}{2}$ gives $t = \frac{\ln 2}{\ln(4/3)}$.
5	(b)	$x_1 = 3x_3$, $x_2 = 4x_4$ is a homogeneous linear system; the others miss $\mathbf{0}$ or fail addition.
6	(a)	Reduce: $x_2 = -4s - 3t$, $x_1 = -5s + t$ with $x_3 = s$, $x_4 = t$.
7	(a)	$v = \frac{1}{2} - \frac{1}{2(t+1)^2}$, $x = \frac{t}{2} + \frac{1}{2(t+1)} - \frac{1}{2}$, so $x(1) = \frac{1}{4}$.
8	(e)	$5r^2 - 6r - 1 = 0$; product of roots $= \frac{-1}{5}$.
9	(a)	$\frac{dy}{y^2} = (\sin x - \cos x)dx$ gives $\frac{1}{y} = \sin x + \cos x - C$; $C = -2$.
10	(e)	$M_y = kx$, $N_x = 2x$, so $k = 2$.
11	(e)	$F = x + e^{xy} + g(y)$ and $g' = 2y$, so $x + e^{xy} + y^2 = c$.
12	(e)	$v = y/x$: $xv' = \frac{4+v^2}{2v}$, so $4 + v^2 = cx$, i.e. $y^2 + 4x^2 = cx^3$.
13	(e)	$\rho = x^{-3/2}$: $x^{-3/2}y = 3x^{3/2} + c$.
14	(c)	$c = \frac{3}{2}$; $y = \frac{1}{3/2-x}$ breaks at $\frac{3}{2}$, and $1 < \frac{3}{2}$.
15	(b)	$v = y^3$, $v' = 3y^2y'$: $xv' = 3x^4 + v$, so $v' - \frac{1}{x}v = 3x^3$.